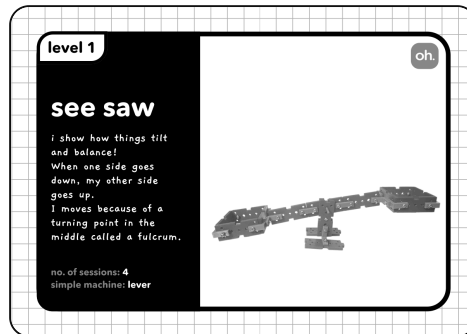


name: _____

level-up exercise

see-saw

robotics · level 1 · mechanics



your challenge

Redesign your see-saw so a lighter side can still lift a heavier one. What do you change about the arms?

parts i used:

☐

☐

☐

my design sketch

 draw & label your new design here

what did i change?

☐

shape

☐

size

☐

parts

☐

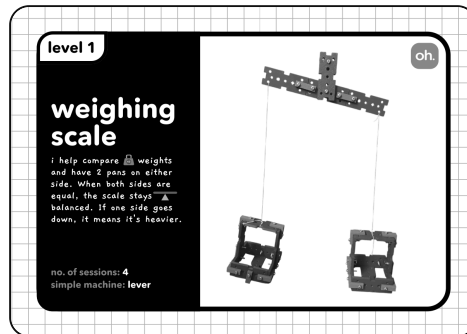
design

name: _____

level-up exercise

weighing scale

robotics · level 1 · mechanics



your challenge

Improve your scale so it can weigh a heavier object without tipping over.

parts i used:

☐

☐

☐

did it improve the model? how?

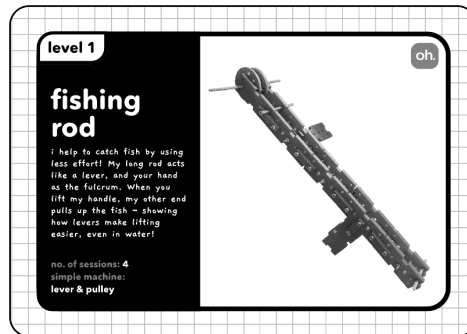
 write what you changed and whether it worked better

name: _____

level-up exercise

fishing rod

robotics · level 1 · mechanics



your challenge

Make your fishing rod lift its catch higher and hold it steady while you reel it in.

parts i used:

☐

☐

☐

did it improve the model? how?

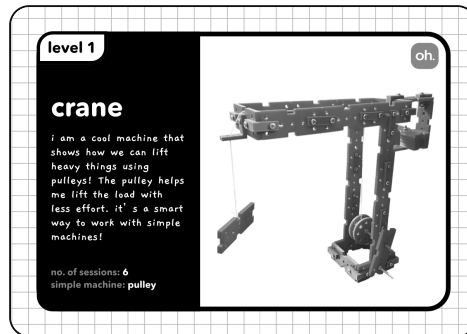
 write what you changed and whether it worked better

name: _____

level-up exercise

crane

robotics · level 1 · mechanics



your challenge

Redesign the crane with a completely different structure while keeping its lifting function.

parts i used:

☐

☐

☐

my design sketch

 draw & label your new design here

what did i change?

☐

shape

☐

size

☐

parts

☐

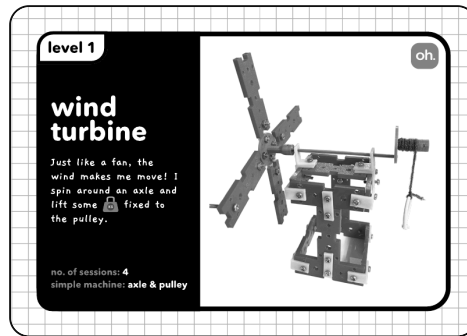
design

name: _____

level-up exercise

wind turbine

robotics · level 1 · mechanics



your challenge

Redesign the wind turbine with a different blade shape so it spins from a gentler push, while keeping the tower stable.

parts i used:

☐

☐

☐

my design sketch

 draw & label your new design here

what did i change?

☐

shape

☐

size

☐

parts

☐

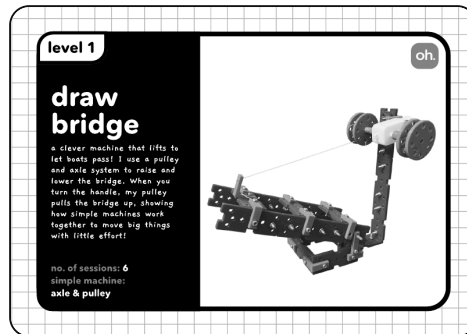
design

name: _____

level-up exercise

drawbridge

robotics · level 1 · mechanics



your challenge

Make the drawbridge open and close smoothly and stop level with the road.

parts i used:

☐

☐

☐

did it improve the model? how?

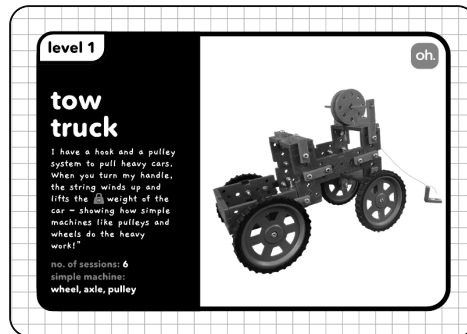
 write what you changed and whether it worked better

name: _____

level-up exercise

tow truck

robotics · level 1 · mechanics



your challenge

Give your tow truck a way to lift a heavier load, or a hook that grips its load better.

parts i used:

☐

☐

☐

did it improve the model? how?

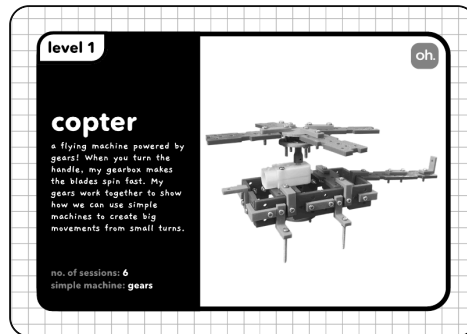
 write what you changed and whether it worked better

name: _____

level-up exercise

copter

robotics · level 1 · mechanics



your challenge

Change the gears so the blades spin faster from the same hand-crank.

parts i used:

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☐

☐

did it improve the model? how?

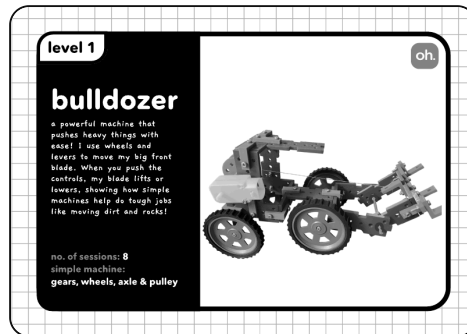
 write what you changed and whether it worked better

name: _____

level-up exercise

bulldozer

robotics · level 1 · mechanics



your challenge

Redesign the bulldozer so its blade can push a bigger pile while it keeps moving steadily.

parts i used:

☐

☐

☐

my design sketch

 draw & label your new design here

what did i change?

☐

shape

☐

size

☐

parts

☐

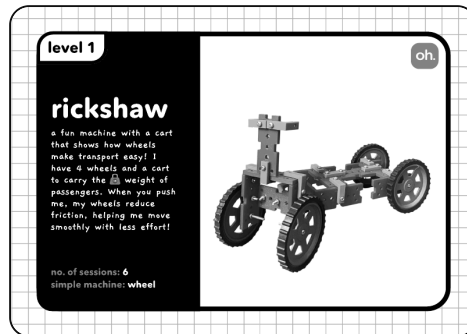
design

name: _____

level-up exercise

rickshaw

robotics · level 1 · mechanics



your challenge

Redesign your rickshaw so it rolls straighter and further from a single push.

parts i used:

☐

☐

☐

my design sketch

 draw & label your new design here

what did i change?

☐

shape

☐

size

☐

parts

☐

design